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Stochastic Broadcast for VANET

Michael Slavik, Imad Mahgoub
 Department of Computer Science and Engineering
 Florida Atlantic University
 Boca Raton, Florida 33431
 Email: mslavik@fau.edu

Abstract—Vehicular Ad-hoc NETWORKS (VANETs) are an emerging important type of wireless ad-hoc network. Unlike many other types of MANETs, VANET applications such as traffic data dissemination are inherently broadcast oriented and require communication protocols to be *anonymous* and *scalable*. Simple broadcast flooding satisfies these requirements, but its performance is highly dependent on network density and may lead to the broadcast storm problem.

This work is the first we are aware of to propose stochastic broadcast as a solution for VANET. Stochastic broadcast instructs nodes to rebroadcast messages according to a retransmit probability. Such a scheme is private since node identification is unnecessary, however it has an undesirable dependency on vehicle density in the same manner as simple flooding. To solve this problem, we demonstrate the link between the mathematical science of continuum percolation and stochastic broadcast. Specifically, that the critical percolation threshold in continuum percolation (≈ 4.5 expected neighbors) translates to the wireless broadcast context. Then we show that nodes can tune the performance of the broadcast system to efficient levels by adjusting the retransmit probability so the apparent density of the network approaches the critical threshold.

I. INTRODUCTION

Vehicular Ad-hoc NETWORKS (VANETs) are a class of wireless networks in which mobile vehicles communicate with each other in an ad-hoc fashion. With 250 million registered vehicles in the U.S. alone in 2006 [1], the potential impact of successfully implementing a national VANET is great and research in the area is active.

VANETs are a special case of MANETs (Mobile Ad-hoc NETWORKS) but have some characteristics not shared by all other types of MANETs. VANET communication protocols must be scalable since a VANET may include huge numbers of members, and for many applications they must be private, since users will be unlikely to adopt technology that allows third parties to monitor their movement. VANETs are highly dynamic, with nodes entering and exiting the network continuously and links between nodes frequently created and destroyed.

However, clearly VANETs share a similar structure to MANETs so work can be adapted from that area. Routing data in ad-hoc networks has been an active area of research for many years, including detailed analysis using graph theory and producing practical protocols such as AODV [2] and DSR [3].

MANET routing protocols often include the need to broadcast information to the entire network. Blindly retransmitting

broadcast packets leads to an exponential growth over time in the number of packets generated by a source broadcast, provided the network is sufficiently dense. Even when retransmission is stable and the broadcast dies out over time, initial bursts of activity often overwhelm networks with collisions and other problems, making the scheme impractical. This is commonly referred to as the *broadcast storm* problem, and is a staple of pure flooding.

Several techniques have been proposed to reduce the number of broadcasts from pure flooding. These can be categorized as counter-based, distance-based, location-based, cluster-based, and stochastic [4]. Counter-based schemes limit the number of times a packet is rebroadcast. The distance-based method considers received signal strength when deciding to rebroadcast, with the goal of only rebroadcasting at the border nodes. Location-based considers the amount of additional area a rebroadcast will cover and cluster-based imposes a topological structure on the nodes and directs only certain nodes or clusters to retransmit broadcasts. Finally, stochastic schemes rebroadcast packets according to some probability. Practical broadcast protocols are generally a hybrid of these fundamental types.

A. Broadcast in VANETs vs MANETs

VANET applications such as traffic data dissemination use broadcast exclusively as the means of communication. Hence solving the broadcast storm problem in an efficient way is the key to developing practical routing protocols for VANETs. A broadcast protocol for VANET must have the following properties.

- **Anonymous:** if node identity is not private, drivers will be reluctant to adopt the technology because information including their location and speed will be traceable
- **Scalable:** a VANET potentially includes millions of nodes and covers entire continents

These assumptions differ from the typical requirements of a MANET broadcast protocol.

1) *Anonymous Data:* There is a large body of elegant work using graph theory to develop efficient broadcast protocols for the MANET context that use k -hop neighbor information to build *connected dominating sets* and multi-point relaying networks (Ex. [5] [6]). However these methods rely heavily on node identification and will be difficult to import to the VANET context given the anonymous data requirement.

Without these cluster-based [4] algorithms, location, distance, counter, and stochastic techniques remain. These generally allow privacy, though practical broadcast protocols already developed using these methods often require node identification [7] [8] [9] [10].

Data privacy implies messages do not contain unique identifiers, so nodes cannot maintain neighbor lists, directly calculate node density, use meaningful packet sequence numbers, etc. The challenge for VANET researchers is to develop broadcast protocols without using these common techniques.

II. RELATED WORK

VANET is a blossoming research area. Important topics include physical layer techniques, data aggregation methods, simulation, applications, and the topic of this paper: data dissemination.

Data dissemination includes peer-to-peer communications and broadcast communications. The peer-to-peer paradigm is the classic networking problem and in VANET is often interested in using vehicles to propagate larger networks such as the Internet. This research typically is an extension of previous MANET work. In contrast broadcast is a one-to-many scheme designed to support applications such as traffic mapping and safety.

The primary problem in wireless broadcast is the *broadcast storm* problem, which is the explosive growth of retransmitted messages that can congest a network to the point it is non-functional.

Researchers have mostly approached this problem in the context of MANET, which uses broadcast as a component of larger protocols (ex. route request in AODV [2]). The work by Ni, et al [4] seems to be often cited as a fundamental work in the area of broadcast storm alleviation. Those authors classify techniques as stochastic, counter, distance, location, and cluster based. Example subsequent work includes [11] [12] [7] [8] (stochastic), [13] [14] (distance), [15] (location), and [16] [5] [6] (cluster). Many of these works contain elements from more than one of the above techniques.

One problem with importing this work directly to VANET is that the problem definition in MANET does not include a privacy constraint. This means that messaging propagates node identifiers along with data, which is unacceptable in VANET.

With this in mind, research in VANET broadcast can be criticized as either violating the privacy constraint or being too simple to be practical. For example, Bako, et al [17] cite the work of Ni et al and others working on broadcast in the MANET context and borrow the concepts of location and stochastic broadcast filtering into VANET, bringing with them their non-private nature. Jiang et al [18] similarly imports cluster-based techniques to VANET that are not private.

We are unaware of any research linking stochastic broadcast, which has properties of anonymity and scalability, and VANET. While the connection between broadcast and percolation has been suggested by others [19], previous work also does not explore the analytical link between stochastic broadcast and continuum percolation. Most importantly, no

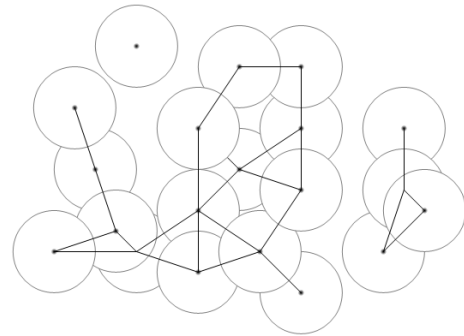


Fig. 1. Sample Continuum Percolation

mention has been given to recognizing the critical density for long distance propagation and the ability to use rebroadcast probabilities to tune the network to the optimal density for efficient data dissemination.

III. CONTINUUM PERCOLATION

Percolation is the study of random flow through a graph. Percolation structures are either *site* or *bond* oriented. In site percolation, vertices (sites) in a lattice are *open* or *closed* according to a Bernoulli process. Open sites that share an edge are connected. In bond percolation, edges (bonds) in a lattice are open or closed according to the same process. In either case, the system is said to *percolate* if the structure is connected with high probability. A phase transition exists at a critical probability p_c between a system that percolates and one that does not.

The percolation threshold is difficult to find in non-trivial cases but has nonetheless been analytically determined in many cases. See Bollobas and Riordan [20] for a detailed treatment. The special case most pertinent to stochastic flooding is called continuum percolation. Here, nodes are placed in a field according to a Poisson point process with some intensity ρ . In any region of area A , the expected number of nodes is ρA . Nodes within a distance of r are then connected together (Figure 1). When the nodes form a single infinite cluster, the system is said to percolate.

Throughout this paper, we assume homogeneous transceivers with transmission radius r uniformly distributed in the plane with intensity ρ . We use the notation $\lambda = \rho\pi r^2$ to denote the expected number of neighbors for each node.

From analytical work in continuum percolation it is known that there exists a critical density (λ_c) below which the network almost surely will be disconnected and above which will likely form an infinite connected cluster. No proof exists for the exact value of λ_c in continuum percolation, but it has been estimated using numeric methods to be between 4.508 and 4.515 with 99.99% confidence [21]. This value is verified in results presented in this paper and will prove to be very important.

IV. STOCHASTIC BROADCAST

In a pure flooding broadcast scheme, all nodes retransmit messages when they receive them. This algorithm is simple to implement and maximizes the likelihood of a message propagating through a network. However, it also can create a broadcast storm in which the medium is suffocated into uselessness. Pure flooding is analogous to continuum percolation.

Stochastic broadcast directs each node to rebroadcast received messages with some probability. This probability may be fixed for all nodes or determined during runtime based on factors such distance from the last hop neighbor, distance from the broadcast origination, etc. No node identifiers are required to accomplish this so the scheme is private, and since all nodes make decisions independently using local information, it is scalable.

When nodes do not always rebroadcast a message, as in stochastic broadcast, the system is not precisely continuum percolation. In a stochastic broadcast system, if a node elects not to participate in the broadcast, it is the same as if the node were not there at all. Conceptually, this implies a stochastic broadcast network with constant retransmit probability should behave the same as a pure flooding network except the density is reduced by the stochastic process. Results presented below assess the validity of this claim.

In this paper, we look at two variations of stochastic broadcast. First, all nodes are given a uniform and constant retransmission probability. Second, the retransmission probability is dynamically determined based on the distance between the receive node and the last hop (nodes that are farther from their last hop have a higher probability of retransmission).

V. SIMULATION RESULTS

The results in this section are found using simulation developed in Java for the purpose of this work. The simulation randomly distributes a given number of nodes in a field sized 1000 x 1000. One node is placed in the center that originates a broadcast. All nodes are homogeneous with transmission radius 10 and the medium is collision-free. Each result presented is an average of 300 simulation trials.

A. Constant Retransmit Probability

The relevant problem is to find the most efficient method of propagating anonymous broadcast messages over a long distance. Consider first a simple stochastic broadcast with constant retransmission probability, including the case of pure flooding ($p = 1.0$). Figures 2 and 3 show the reachability at a distant node and the total number of transmitted messages plotted against the density of nodes in the case of $p = 1.0$.

As predicted by the continuum percolation analysis, a phase transition is exhibited around $\lambda \approx 4.5$. One can see the broadcast storm begin to occur above this threshold. Like so many other problems in engineering, a trade off is shown here between reachability and bandwidth utilization. The goal is to minimize bandwidth subject to a reachability constraint.

The obvious problem is that in VANET or most other applications system designers cannot choose λ . The density is

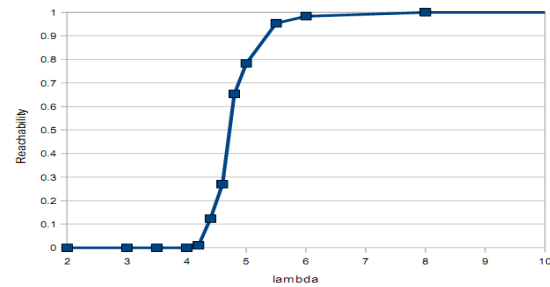


Fig. 2. Reachability vs λ for $p = 1.0$

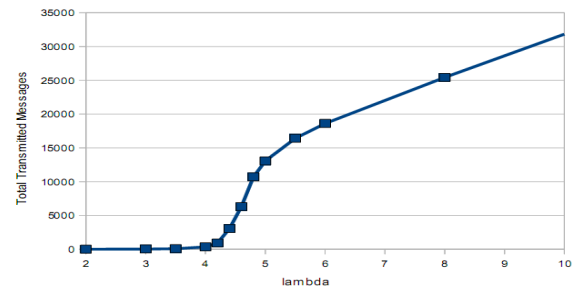


Fig. 3. Number of messages transmitted vs λ for $p = 1.0$

a stochastic property that is constantly changing. Thus pure flooding will exhibit poor performance in all cases except when λ is near the critical threshold.

Stochastic flooding offers a solution. Assuming nodes have the ability to estimate the surrounding density, they should be able to reduce their rebroadcast probability in order to tune the apparent density to a lower value. For example, a network has density $\lambda = 10$. The retransmit probability could then be set to 0.5, so that only half the nodes in the network will participate in the flood, thus reducing the apparent density of the network to an efficient value of $\lambda = 5$.

Figures 4 and 5 compare the performance of stochastic flooding to pure flooding, this time with independent axis as the apparent density λp where p is the constant retransmit probability.

Notice the important result: it makes little difference what value of p is used. Rather, the apparent density λp is the primary factor in determining performance. Thus nodes can adjust the performance of broadcasting during runtime by

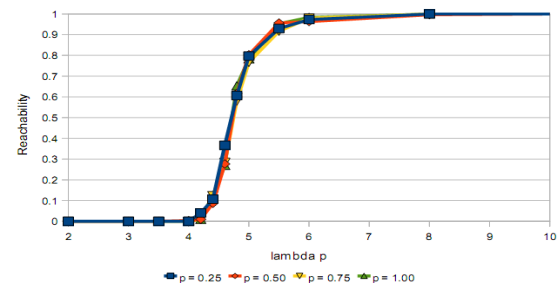


Fig. 4. Reachability vs λp for constant p

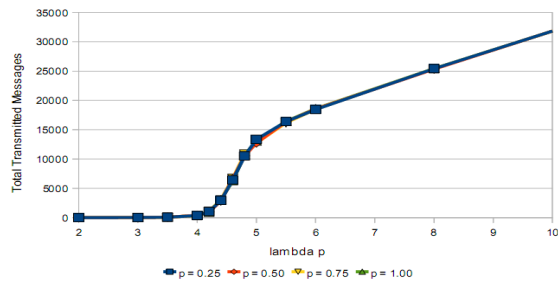


Fig. 5. Number of messages transmitted vs λp for constant p

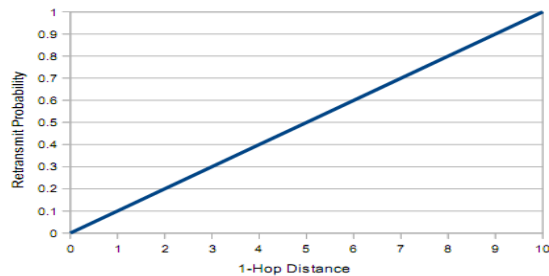


Fig. 6. Retransmit probability in Distance method

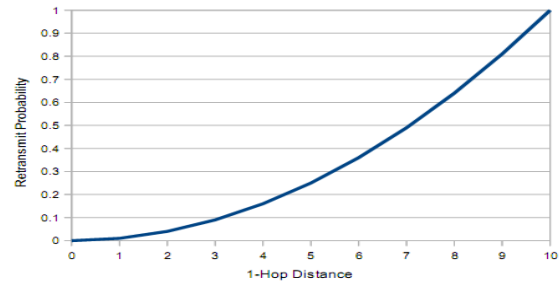


Fig. 7. Retransmit probability in Distance² method

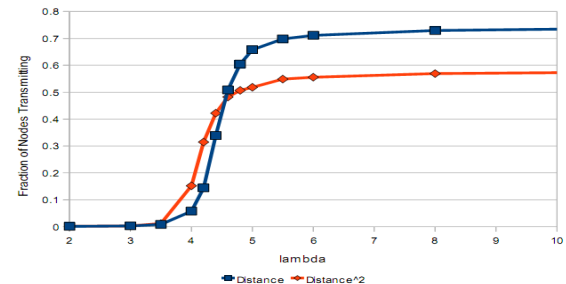


Fig. 8. Apparent retransmit probability for Distance methods

adjusting the retransmit probability and thus the apparent density of the network.

Density is a crucial factor in the performance of a broadcast system and is entirely out of the control of VANET designers. The ability to control apparent density using stochastic broadcast makes broadcast possible *without* violating design constraints of anonymity and scalability.

B. Retransmit Probability by Distance

Next consider a system with dynamic retransmit probability related to the distance between the receiving node and transmitting node (1-hop distance). Using 1-hop distance as a determining factor is a common technique in broadcast schemes because it appeals to the intuition that if only nodes that are far apart participate in a broadcast, there is likely to be less overlap between broadcasts. If two nodes are positioned close together, little is gained from both of them rebroadcasting a message.

Two variations of this method are considered here. The first is a retransmit probability that is directly proportional to 1-hop distance (Distance method) as shown in Figure 6. The second is a retransmit probability that is proportional to the 1-hop distance squared (Distance² method) as shown in Figure 7.

When examining the results in these cases, it appears the behavior is similar to the case of constant retransmit probability, but it is unclear how to compare them. An interesting property of the constant retransmit probability case is that as the density of the network gets very large, the fraction of nodes in the network that transmit a message approaches the retransmit probability. Assume for a moment that this is the case for the Distance methods as well. Figure 8 shows the fraction of nodes that broadcast versus density in these cases.

This data shows that for the Distance method, the fraction of nodes participating in the broadcast approaches 0.75 as density increases. Here this value is referred to as the apparent retransmit probability (or apparent p). The same value for the Distance² method is about 0.57. Figures 9 and 10 compare the performance of the Distance methods to pure flooding, this time using apparent p to calculate the apparent density λp .

From these results our assumption seems justified. The Distance methods perform similar to constant retransmit probability systems with the retransmit probability set to the apparent p . So despite intuition to the contrary, little is gained by basing the retransmit probability on the 1-hop distance. These systems in reality perform about the same as an equivalent constant retransmit probability system.

VI. CONCLUSION

We have proposed stochastic broadcast as an efficient solution to the VANET data dissemination problem. Unlike

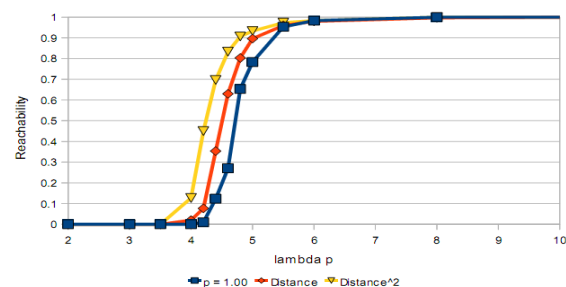


Fig. 9. Reachability vs λp for Distance methods

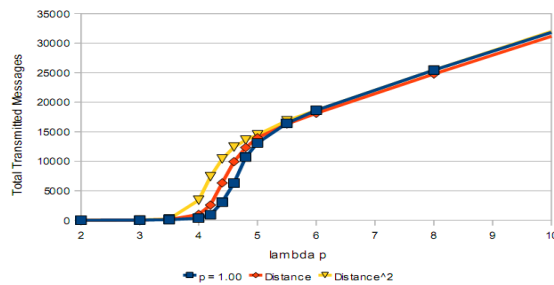


Fig. 10. Number of messages transmitted vs λp for Distance methods

many broadcast schemes from the MANET context, it does not require nodes to identify themselves and thus is private. Anonymity of transmitted data is crucial in VANET, since users and auto-makers will not adopt a technology that allows vehicle movement to be tracked. Furthermore, stochastic broadcast is simple to implement with only local information required on each vehicle and no overhead messaging required. This makes the scheme efficient and scalable, another requisite of any practical VANET communication proposal.

Stochastic broadcast requires that all nodes determine a probability with which they will retransmit received messages. Unfortunately choosing this probability is not a simple task. Choose a value that is too small, and the network may not provide the required reachability. Too large, the system may exhibit the broadcast storm problem in the same manner as simple flooding. Further, the density of vehicles affects the performance of the system, so even if a good value of the retransmit probability is chosen at design time, it will only be efficient for a narrow range of vehicle densities.

Examining this problem, we begin by showing the similarity between continuum percolation and stochastic broadcast. Most importantly, that the density threshold for long range communication in wireless broadcast is the same as the critical density in continuum percolation, known to be about 4.5. When the vehicle density is near the critical threshold, the network provides high reachability with minimum bandwidth utilization.

Finally, we show that nodes can tune the performance of the broadcast system to efficient levels by adjusting the retransmit probability so the apparent density of the network approaches the critical threshold. This discovery allows designers to overcome the distasteful dependency on vehicle density by estimating local vehicle density and setting the retransmit probability to a value so that the network appears to have the desired density. This makes the use of stochastic broadcast in VANET a practical possibility.

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